

WASP-12b

R-0362

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_231114 · created 2026-07-18T13:40:42+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460261.8499 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.097 +/- 0.043
Transit depth (Rp/Rs)^2	0.94 +/- 0.84 [%]
Orbital Inclination (inc)	85.71 +/- 2.88
Airmass coefficient 1 (a1)	0.98 +/- 0.12
Airmass coefficient 2 (a2)	0.013 +/- 0.068
Scatter in the residuals of the lightcurve fit is	11.77 %
Best Comparison Star	#3 - [366, 331]
Optimal Aperture	6.03
Optimal Annulus	7.54
Transit Duration (day)	0.1245 +/- 0.0067

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.097 $\pm$ 0.043 vs 0.1178 (deviation 0.44 σ)
Duration fit vs published	0.1245 d vs 0.125 d (deviation 0.07 σ)
Mid-time O–C	-1569.9 min
Depth SNR	2.1
Residual scatter	11.77 %

Light curve

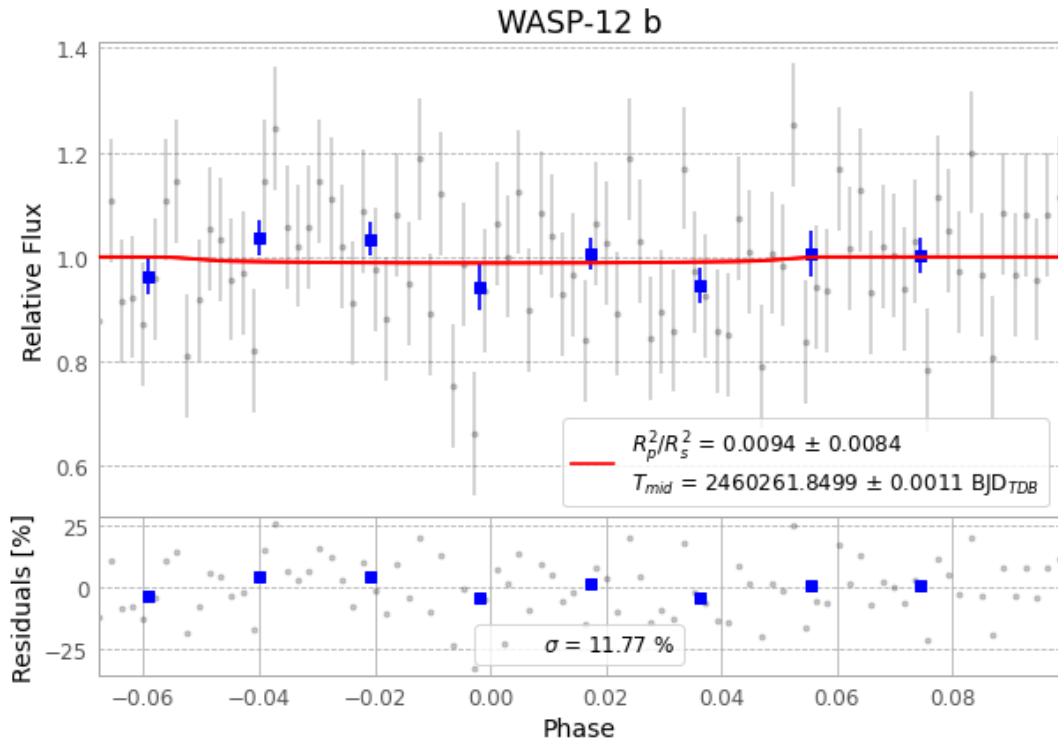


Figure 1 — FinalLightCurve_WASP-12 b_2023-11-12.png (SHA-256 ca7a9694213c9af1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [382, 329], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ce7bc9e7b640a458...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 9e597cf

Data provenance

88 FITS frames (58 MB), WASP-12231113063015.FITS → WASP-12231113105116.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-231114.log (9b8130d5042a...), FinalLightCurve_WASP-12 b_2023-11-12.png (ca7a9694213c...), FinalLightCurve_WASP-12 b_2023-11-12.csv (5ddc32a396b9...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 13:40Z.